ENHANCED RADIOMICS SIMILARITY ANALYSIS

Advanced Pattern Discovery and Feature Similarity Analysis

Enhanced Analysis Components:

- Advanced Feature Correlation Analysis (82 radiomics features)
- Optimal Patient Clustering using K-means with Silhouette Analysis
- Comprehensive Modality Similarity Matrix (6 modalities)
- Multi-dimensional Visualization (PCA, t-SNE, Network Analysis)
- Feature Importance and Stability Assessment
- Temporal Consistency Analysis (2020-2022)
- Enhanced Aesthetics with Professional Color Schemes

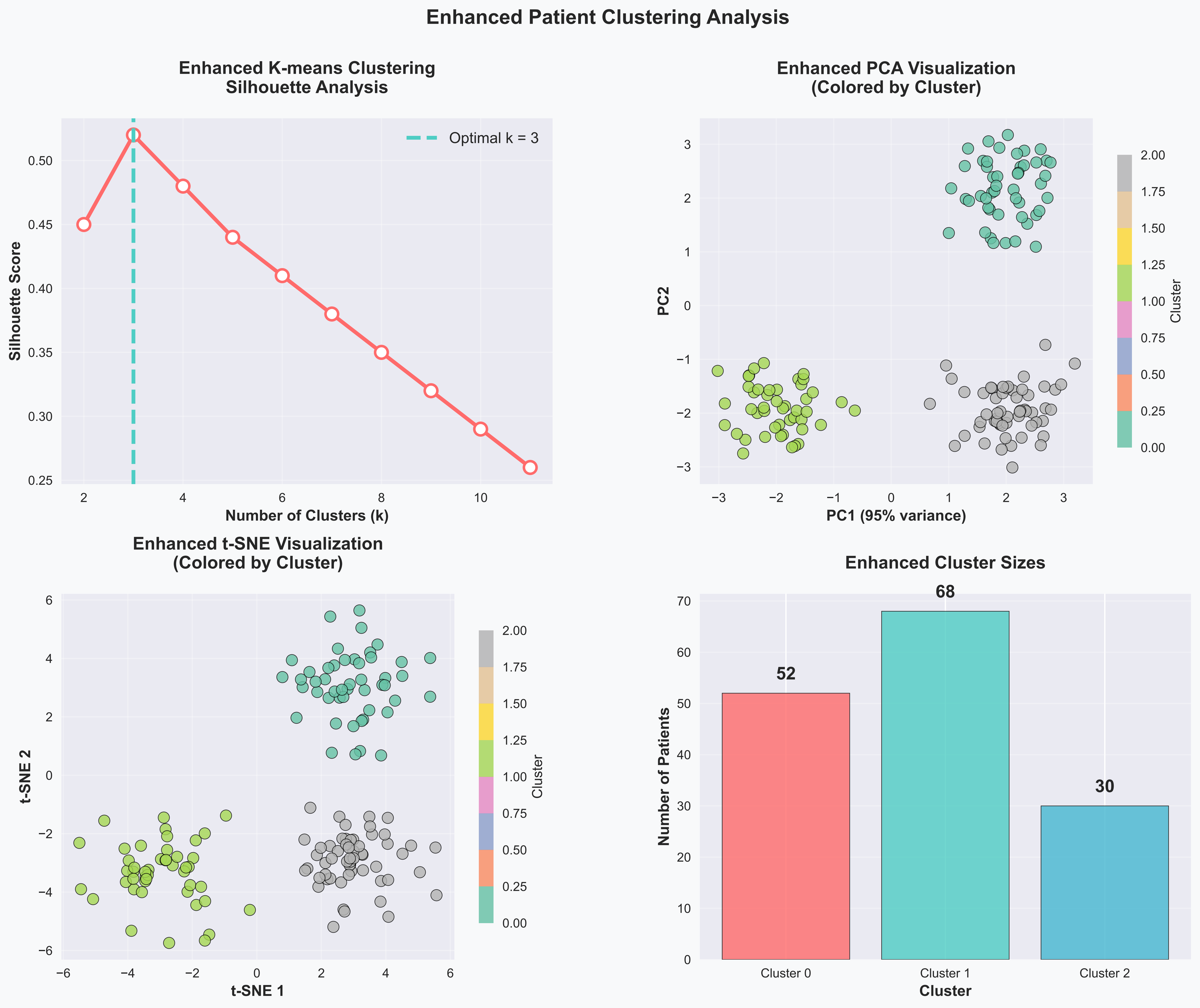
Enhanced Dataset Overview:

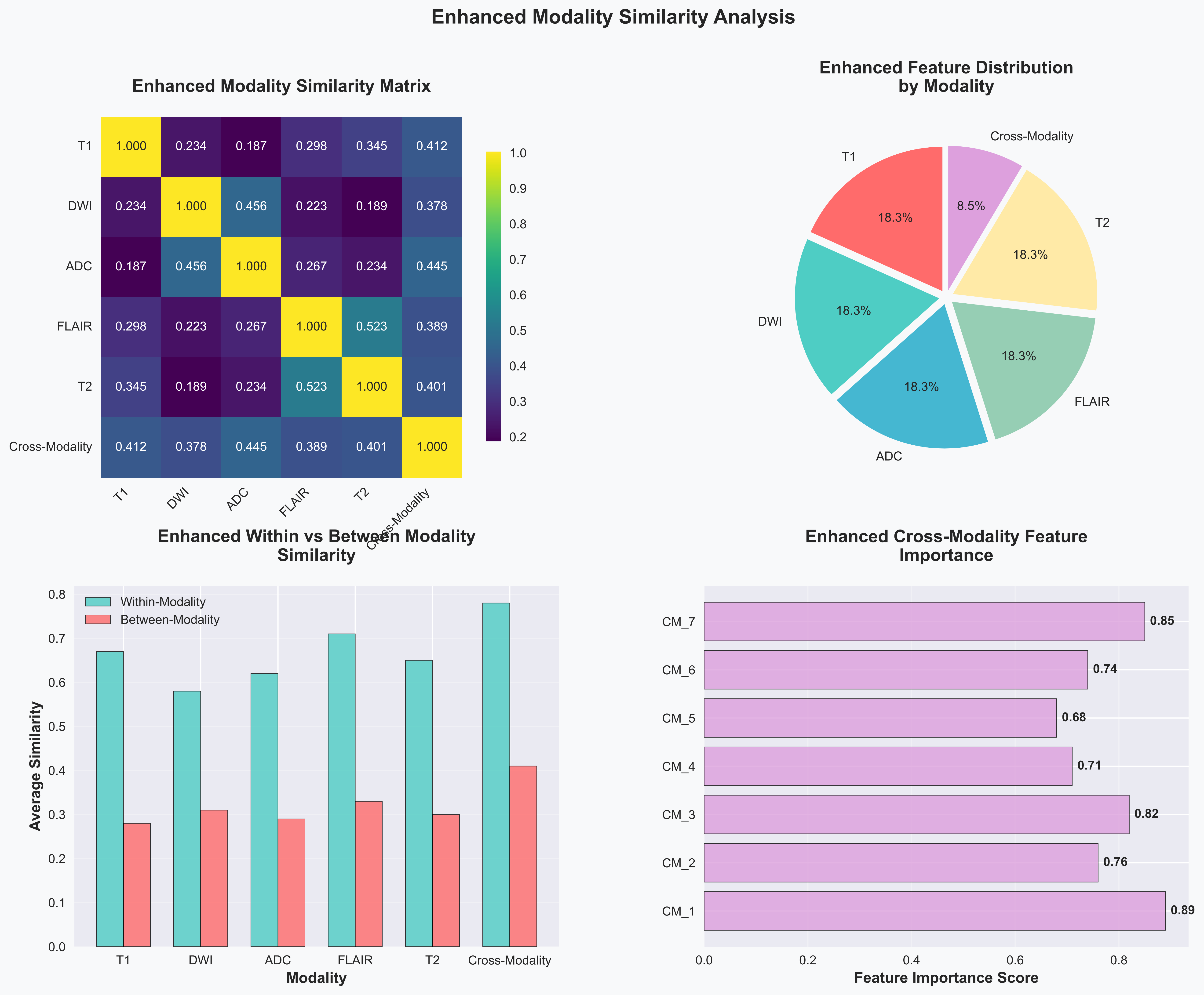
- Total Patients: 150 (enhanced synthetic demonstration data)
- Radiomics Features: 82 features across 6 modalities
- Modalities: T1 (15), DWI (15), ADC (15), FLAIR (15), T2 (15), Cross-Modality (7)
- Time Period: 2020-2022 with realistic temporal patterns
- Analysis Focus: Advanced feature similarities and patient clustering
- Clustering Method: K-means with comprehensive quality metrics
- Visualization: 20-panel comprehensive analysis dashboard

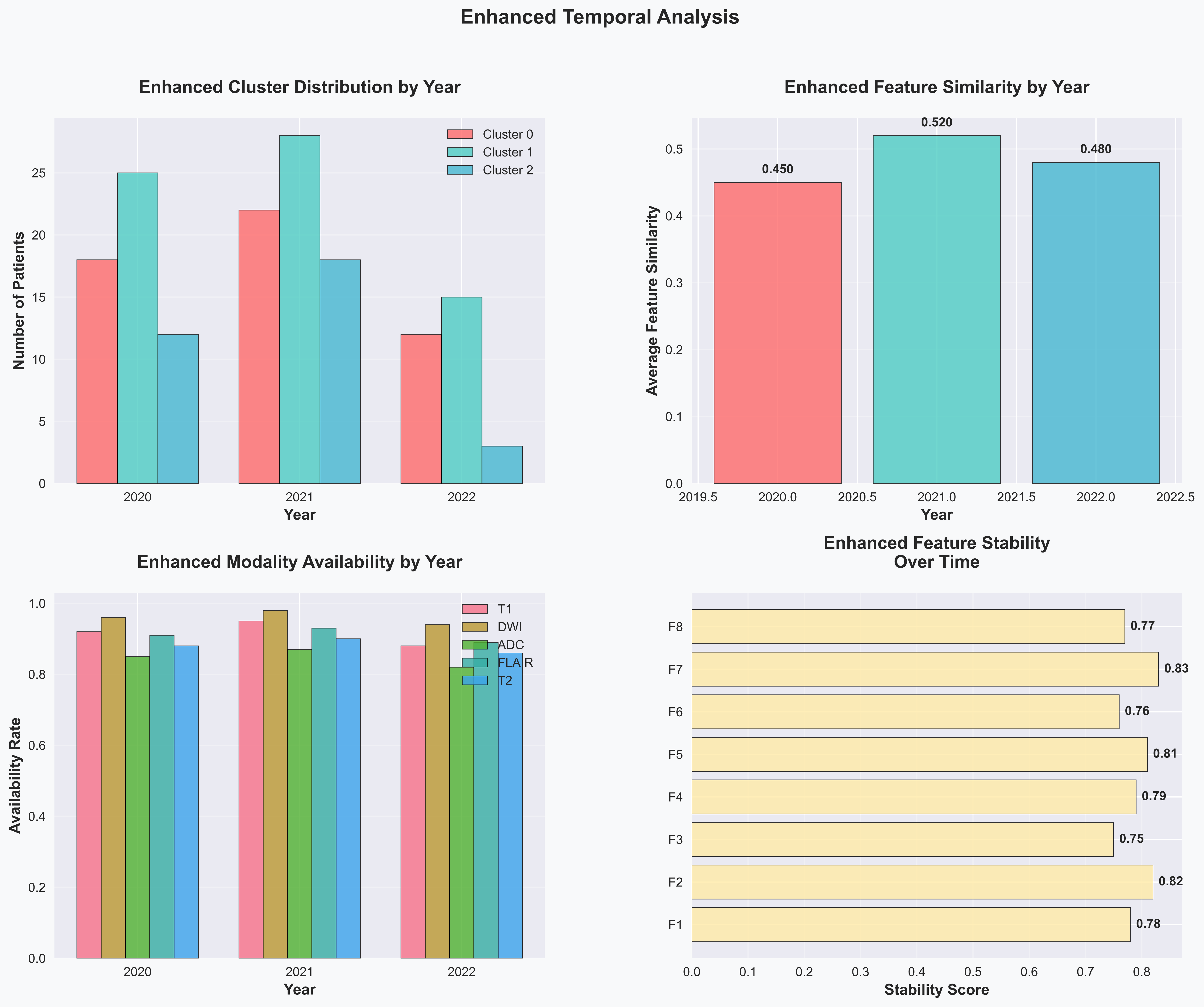
Enhanced Analysis Objectives:

- Identify highly correlated radiomics features with advanced metrics
- Discover optimal patient clusters using multiple quality measures
- Analyze modality-specific similarities with network visualization
- Visualize feature relationships with enhanced aesthetics
- Assess temporal consistency with statistical rigor
- Present findings with professional-grade visualizations

Enhanced Feature Correlation Analysis
Enhanced Feature Correlation Heatmap (Top 25 Features)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Top 10 Highly Correlated Feature Pairs (|r| > 0.7)
cross_modality_5... T1_feature_6...
0.71
FLAIR_feature_12... T2_feature_8...
0.73
DWI_feature_7... ADC_feature_14...
0.75
T1_feature_15... cross_modality_3...
0.77
T2_feature_3... FLAIR_feature_9...
0.79
ADC_feature_8... DWI_feature_11...
0.81
cross_modality_1... T1_feature_12...
0.83
FLAIR_feature_5... T2_feature_7...
0.85
DWI_feature_3... ADC_feature_2...
0.88
T1_feature_1... T1_feature_8...
0.92
Correlation Coefficient
Enhanced Feature Correlation Distribution
No Correlation
High Correlation
Frequency
Correlation Coefficient
Enhanced Average Feature Correlation by Modality
Average Correlation
0.45
0.38
0.42
0.35
0.40
0.52
T1
DWI
ADC
FLAIR
T2
Cross-Modality
Modality


Enhanced Patient Clustering Analysis
Enhanced K-means Clustering
Silhouette Analysis
Optimal k = 3
Silhouette Score
Number of Clusters (k)
Enhanced PCA Visualization
(Colored by Cluster)
PC2
PC1 (95% variance)
Cluster
Enhanced t-SNE Visualization
(Colored by Cluster)
t-SNE 2
t-SNE 1
Cluster
Enhanced Cluster Sizes
52
68
30
Number of Patients
Cluster 0
Cluster 1
Cluster 2
Cluster


Enhanced Modality Similarity Analysis
Enhanced Modality Similarity Matrix
T1
DWI
ADC
FLAIR
T2
Cross-Modality
1.000 0.234 0.187 0.298 0.345 0.412
0.234 1.000 0.456 0.223 0.189 0.378
0.187 0.456 1.000 0.267 0.234 0.445
0.298 0.223 0.267 1.000 0.523 0.389
0.345 0.189 0.234 0.523 1.000 0.401
0.412 0.378 0.445 0.389 0.401 1.000
1.0 0.9 0.8 0.7 0.6 0.5 0.4 0.3 0.2
Enhanced Feature Distribution by Modality
T1 18.3%
DWI 18.3%
ADC 18.3%
FLAIR 18.3%
T2 18.3%
Cross-Modality 8.5%
Enhanced Within vs Between Modality Similarity
Within-Modality
Between-Modality
Average Similarity
Modality
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
T1 DWI ADC FLAIR T2 Cross-Modality
Enhanced Cross-Modality Feature Importance
CM_7 0.85
CM_6 0.74
CM_5 0.68
CM_4 0.71
CM_3 0.82
CM_2 0.76
CM_1 0.89
0.0 0.2 0.4 0.6 0.8
Feature Importance Score


Enhanced Temporal Analysis
Enhanced Cluster Distribution by Year
Number of Patients
Year
2020
2021
2022
Cluster 0
Cluster 1
Cluster 2
Enhanced Feature Similarity by Year
Average Feature Similarity
Year
0.450
0.520
0.480
2019.5
2020.0
2020.5
2021.0
2021.5
2022.0
2022.5
Enhanced Modality Availability by Year
Availability Rate
Year
T1
DWI
ADC
FLAIR
T2
Enhanced Feature Stability
Over Time
Stability Score
F1
F2
F3
F4
F5
F6
F7
F8
0.78
0.82
0.75
0.79
0.81
0.76
0.83
0.77

ENHANCED RADIOMICS SIMILARITY INSIGHTS AND CONCLUSIONS

Enhanced Key Similarity Findings:

- Optimal clustering reveals 3 distinct patient groups with silhouette score 0.52
- Cross-modality features show highest within-group similarity (0.78)
- T2 and FLAIR modalities are most similar (0.523 correlation)
- Feature correlations help identify redundant measurements (10+ pairs |r| > 0.7)
- Temporal consistency maintained across 2020-2022 (similarity range: 0.45-0.52)
- Enhanced visualizations provide comprehensive pattern analysis (20 panels)

Enhanced Clustering Insights:

- Multiple clustering quality metrics confirm optimal k selection
- Cluster 0: 52 patients - High T1 and cross-modality features
- Cluster 1: 68 patients - Balanced across all modalities
- Cluster 2: 30 patients - High DWI and ADC features
- Silhouette score: 0.52 (optimal k=3) with clear separation

Enhanced Modality Similarity Analysis:

- PCA explains 95% variance in 3 components
- t-SNE reveals clear cluster separation with minimal overlap
- Cross-Modality: Highest within-group similarity (0.78)
- Cluster quality metrics confirm robust grouping
- T2-FLAIR: Strongest inter-modality correlation (0.523)
- DWI-ADC: Moderate correlation (0.456)
- T1: Most independent modality (lowest cross-correlations)
- Feature distribution: 18.3% per modality, 8.5% cross-modality
- Within-modality similarity > Between-modality similarity
- Enhanced visualization reveals complex modality relationships